UNDER FINITE PRESENTATION THE FINITE TARGET IN THE ZARISKI MAIN THEOREM FOR SPACES CAN BE CHOSEN OF FINITE PRESENTATION

THE STACKS PROJECT AUTHORS

Editorial note. Statement span is lines 7805-7812 of sources/stacks-spaces-more-morphisms/pinned-original-source.tex; proof span is lines 7814-7827, the complete author proof (12 body lines).

Editorial note. Navigation stubs: 4 cited-but-not-included labels are supplied as non-load-bearing pointers with their pinned tags where the snapshot tag index has them.

Editorial note. extra_packages is empty on purpose: the retained preamble already loads xy with options and a second declaration clashes in the assembler wrapper.

Editorial note. Text is retained verbatim, including the author's own typos and placeholders.

Editorial note. Counted B side; schemes twin (More on Morphisms zariski-main-theorem addendum) not in this pool.

Allowed standard prerequisites (editorial).

- Explicit prerequisite boundary. In-source lemma-quasi-finite-separated-pass-through-finite (published card S3-spaces-zariski-main-theorem, L16211); Limits of Spaces, lemmas finite-in-finite-and-finite-presentation, descend-opens, finite-type-eventually-closed.

Source reference. Original lemma 05K0: sources/stacks-spaces-more-morphisms/pinned-original-source.tex, source lines 7755–7768. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 0827: sources/stacks-spaces-more-morphisms/pinned-original-source.tex, source lines 7827–7827. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 07VR: sources/stacks-spaces-more-morphisms/pinned-original-source.tex, source lines 7827–7827. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 0828: sources/stacks-spaces-more-morphisms/pinned-original-source.tex, source lines 7827–7827. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Date: stacks-project:spaces-more-morphisms.tex:local-snapshot-20261011; GNU FDL 1.2 or later.

Lemma 0.1. *With notation and hypotheses as in Lemma 05K0. Assume moreover that f is locally of finite presentation. Then we can choose the factorization such that T is finite and of finite presentation over Y.*

Proof. By Limits of Spaces, Lemma 07VR we can write $T = \lim T_i$ where all T_i are finite and of finite presentation over Y and the transition morphisms $T_{i'} \to T_i$ are closed immersions. By Limits of Spaces, Lemma 0827 there exists an i and an open subscheme $U_i \subset T_i$ whose inverse image in T is X. By Limits of Spaces, Lemma 0828 we see that $X \cong U_i$ for large enough i. Replacing T by T_i finishes the proof. □